

TUTORIAL SHEET

TOPIC:HASHING

Question 1

- a) Define a hash table and explain the role of a hash function.
- b) What is a collision? Explain open addressing as a method to resolve collisions.
- c) Using a hash table of size 10 and $h(k) = k \bmod 10$, insert keys 15, 25, 35, 20 using linear probing and show the final table.

Question 2

Explain separate chaining and how it handles collisions. Insert keys 7, 17, 27, 37 into a hash table of size 10 using $h(k) = k \bmod 10$ with separate chaining. Draw the resulting table.

Question 3

What is double hashing, and why is it better than linear probing in some cases? Explain the concept of rehashing in hash tables. Why is it needed?